



FULL ROADMAP

20 July - 27 Nov 2026

Daily: 2 hrs Tech + 2 hrs Content | 18 Weeks | Zero to Job Ready

📅 20 July - 27 November 2026 | 18 Weeks | Zero to Job Ready

📁 Roadmap Overview

Month	Duration	Theme	Job Readiness (End)
Month 1	20 July - 20 Aug	Python Solid + NLP Practical + LLM Basics	15%
Month 2	20 Aug - 20 Sept	LangChain + RAG Pipeline	40%
Month 3	20 Sept - 20 Oct	Agentic AI — Tools + Multi-step Agents	65%
Month 4	20 Oct - 27 Nov	NLP + Agentic AI Combo + Polish + Interview Ready	80%

🕒 **Daily Commitment:** 2 hrs Tech + 2 hrs Content, every single day, across all 18 weeks

20 July - 20 Aug

Theme: Python Solid + NLP Practical + LLM Basics

📅 WEEK 1 · 20-26 Jul — Python Intermediate

Resource: *freeCodeCamp Python (YouTube) + W3Schools Python*

Tech Topics

- OOP: Classes, Objects, Inheritance, Methods
- File Handling: read, write, append
- Error Handling: try / except / finally
- requests library — API call karo
- List comprehension, Lambda functions, Virtual environments + .env files

DSA (Weekend only)

- Two Sum, Reverse String, FizzBuzz, Palindrome Number, Maximum Subarray

Content

- 2nd channel: 7 shorts batch record (weekend)
- Instagram: 4-5 posts (repurpose shorts)
- Main channel: Research + Script — "Python zero se sikhna shuru kiya"

🎯 Target: 5 problems

✓ Checklist

- ☐ OOP — khud se class likhi
- ☐ API call kiya requests se
- ☐ 5 DSA problems
- ☐ GitHub pe week 1 push
- ☐ 7 shorts batch recorded

📅 WEEK 2 · 27 Jul - 2 Aug — NLP Practical (Zero se)

Resource: *HuggingFace Docs (huggingface.co/docs) + Krish Naik YouTube — "NLP Playlist"*

Tech Topics

- NLP kya hai — Tokenization, Stemming, Lemmatization (concept)
- HuggingFace transformers install karo
- pipeline use karna — sentiment analysis, NER, text classification
- Embeddings practically karo — sentence-transformers library
- Word2Vec concept — similar words find karo
- TF-IDF practically — `sklearn.feature_extraction.text`
- Khud ek simple sentiment classifier banao (movie reviews pe)

DSA (Weekend only)

- Valid Parentheses, Merge Two Sorted Lists, Best Time to Buy Stock, Contains Duplicate, Single Number

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: "Python zero se sikhna" — shoot + edit + **upload** 📺

🎯 Target: 5 problems

✓ Checklist

- ☐ HuggingFace pipeline se sentiment analysis kiya
- ☐ Simple sentiment classifier banaya
- ☐ TF-IDF + Word2Vec concept practically try kiya
- ☐ 5 DSA problems
- ☐ Main channel video 1 uploaded

📅 WEEK 3 · 3-9 Aug — ABSA Own Karo + Python Advanced

Resource: *ABSA code khud padhna + Real Python — OOP articles*

Tech Topics

- ABSA code line by line padhna — data loading, preprocessing, BERT model, BIO heads, span extraction
- Har part ka diagram paper pe draw karo
- ABSA ko khud explain karo — voice record karo (5 min)
- Python Advanced: Decorators, Generators, Iterators
- GitHub pe ABSA properly upload karo with detailed README

DSA (Weekend only)

- Climbing Stairs, House Robber, Power of Two, Missing Number, Move Zeroes

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: Script likhna — "NLP kya hai — simple Hindi explanation"

🎯 Target: 5 problems

✓ Checklist

- ☐ ABSA code samjha — bina notes ke explain kar sako
- ☐ ABSA GitHub pe README ke saath uploaded
- ☐ Python advanced concepts clear
- ☐ 5 DSA problems
- ☐ Next video script ready

📅 WEEK 4 · 10-20 Aug — OpenAI/Gemini API + FastAPI Basics

Resource: *Google AI Studio (free) + FastAPI Official Docs Chapter 1-6*

Tech Topics

- API key setup, .env mein store karna
- Prompt Engineering: System prompt, User prompt, Temperature, max_tokens
- CLI Chatbot — loop mein user input, API call, conversation history
- FastAPI install — GET / pehla endpoint
- Path params + Query params + Request Body + Pydantic models
- Chatbot ko FastAPI pe move karo — /chat POST endpoint
- CORS middleware + Deploy to Render.com (free)

DSA (Weekend only)

- Linked List Cycle, Middle of Linked List, Reverse Linked List, Binary Search, First Bad Version

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: "NLP kya hai" — shoot + edit + **upload** 📤

🎯 Target: 5 problems

✓ Checklist

- ☐ CLI + FastAPI chatbot live on Render
- ☐ Postman se test kiya
- ☐ Prompt engineering basics samjha
- ☐ 5 DSA problems
- ☐ Main channel video 2 uploaded

📊 Month 1 End

- Python — Intermediate ✓
- NLP Practical — Basic-Intermediate ✓
- ABSA — Owned + Explained ✓
- LLM API — Basic usage ✓
- FastAPI — Basic ✓
- DSA — ~20 problems ✓
- Live projects — 1 (chatbot) ✓
- Main channel videos — 2 ✓
- **Job Readiness — 15%**

20 Aug - 20 Sept

Theme: LangChain + RAG Pipeline

📅 WEEK 5 · 20-26 Aug — LangChain Basics

Resource: *LangChain Official Docs* — "Get Started" + *Analytics Vidhya LangChain Guide*

Tech Topics

- ChatGoogleGenerativeAI / ChatOpenAI setup
- PromptTemplate + ChatPromptTemplate
- LCEL pipe operator | — prompt | llm chain
- ConversationBufferMemory — memory wala chatbot
- Output parsers: StrOutputParser, JsonOutputParser

DSA (Weekend only)

- Maximum Depth of Binary Tree, Symmetric Tree, Invert Binary Tree, Path Sum, Count Complete Tree Nodes

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: Script — "LangChain kya hai — simple Hindi mein"

🎯 **Target: 5 problems**

✓ Checklist

- ☐ LangChain memory chatbot working
- ☐ LCEL pipe syntax samjha
- ☐ 5 DSA problems
- ☐ Script ready

📅 WEEK 6 · 27 Aug - 2 Sept — Vector DB + Embeddings

Resource: *ChromaDB Docs* + *LangChain Docs* "Vector Stores" section

Tech Topics

- Embeddings — Week 2 ka sentence-transformers ab LangChain ke saath use karo
- ChromaDB install + setup
- Documents add karo, similarity search karo
- LangChain + Chroma integration
- Document loaders: TextLoader, PyPDFLoader
- RecursiveCharacterTextSplitter — chunking

DSA (Weekend only)

- Number of Islands, Flood Fill, Clone Graph, Course Schedule, 01 Matrix

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: "LangChain kya hai" — shoot + edit + **upload** 📺

🎯 **Target: 5 problems**

✓ Checklist

- ☐ ChromaDB similarity search working
- ☐ PDF load + chunk kiya
- ☐ Week 2 embeddings + ChromaDB connect ho gaya — concept clear
- ☐ 5 DSA problems
- ☐ Main channel video 3 uploaded

📅 WEEK 7 · 3-9 Sept — RAG Pipeline

Resource: LangChain Docs "RAG" section + Krish Naik YouTube "RAG Tutorial"

Tech Topics

- RAG flow: Load → Split → Embed → Store → Retrieve → Generate
- `create_retrieval_chain` — proper RAG chain
- FastAPI integrate karo — `/ask` POST endpoint
- Simple HTML UI — file upload + chat interface
- Deploy to Render

DSA (Weekend only)

- Two Sum II, 3Sum, Container with Most Water, Valid Palindrome, Trapping Rain Water

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: Script — "RAG app banaya apne college notes ke liye"

🎯 Target: 5 problems

✓ Checklist

- ☐ RAG pipeline working end-to-end
- ☐ FastAPI + HTML UI + Deployed
- ☐ 5 DSA problems
- ☐ Script ready

📅 WEEK 8-9 · 10-20 Sept — PROJECT 1 COMPLETE: Study Assistant RAG App

Resource: LangChain Docs advanced retrieval + LangSmith

Tech Topics

- Multiple PDFs support
- MMR retrieval — diversity-based better answers
- `create_history_aware_retriever` — follow-up questions support
- LangSmith setup — free tier, traces dekho
- README: screenshots, demo GIF (Loom se record karo), live URL

DSA (Weekend only)

- 3 problems per weekend = 6 problems

Content

- 2nd channel: 7 shorts/week batch
- Instagram: 4-5 posts/week
- Main channel: "RAG app banaya" — shoot + edit + **upload** (Week 8) 📅
- **LinkedIn pe project post karo**

✓ Checklist

- ☐ PROJECT 1 deployed ✓
- ☐ Multi-PDF + conversation history working
- ☐ Professional README + Demo GIF
- ☐ LinkedIn pe post kiya
- ☐ **Month 2 total DSA: ~36 problems** ✓

📊 Month 2 End

- LangChain — Intermediate ✓
- RAG Pipeline — Can build independently ✓
- ChromaDB — Working knowledge ✓
- NLP → Embeddings bridge — Clear ✓
- DSA — ~36 problems total ✓
- Live Projects — 1 (Study Assistant) ✓
- Main channel videos — 4 total ✓
- **Job Readiness — 40%**

20 Sept - 20 Oct

Theme: Agentic AI — Tools + Multi-step Agents

📅 WEEK 10 · 20-26 Sept — LangChain Agents + Tool Calling

Resource: *LangChain Docs "Agents"* + *JetBrains Blog "LangChain 2026 Tutorial"*

Tech Topics

- ReAct pattern — Think → Act → Observe loop
- Built-in tools: DuckDuckGoSearchRun, WikipediaQueryRun
- Custom tool: @tool decorator
- create_react_agent — agent + multiple tools
- AgentExecutor — run + output
- Tool error handling + graceful fallback

DSA (Weekend only)

- Coin Change, House Robber II, Longest Common Subsequence, Word Break, Jump Game

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: Script — "AI Agent kya hota hai — real example se"

🎯 Target: 5 problems

✓ Checklist

- ☐ ReAct agent 3 tools ke saath working
- ☐ Custom tool banana aaya
- ☐ 5 DSA problems

📅 WEEK 11 · 27 Sept - 3 Oct — LangGraph Basics

Resource: *LangGraph Docs "Get Started"* + *"Conceptual Guide"*

Tech Topics

- LangGraph vs AgentExecutor — kab kya use karein
- StateGraph + TypedDict — state define karna
- Nodes + Edges + Conditional edges
- ToolNode — tools ko LangGraph mein use karna
- Human-in-the-loop — interrupt_before
- LangSmith se traces dekho

DSA (Weekend only)

- Decode Ways, Partition Equal Subset, Target Sum, LRU Cache, Min Stack

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: "AI Agent kya hota hai" — shoot + edit + **upload** 📤

🎯 Target: 5 problems

✓ Checklist

- ☐ LangGraph basic flow samjha
- ☐ Multi-step agent LangGraph pe working
- ☐ 5 DSA problems
- ☐ Main channel video 5 uploaded

Resource: LangChain + LangGraph Docs + PyPDF2

Tech Topics

- Resume parser tool — PDF → structured data (PyPDF2 + LLM)
- JD parser tool — key requirements extract
- Match scorer tool — skills comparison, 0-100% score
- Cover letter generator tool
- LangGraph workflow — sabhi tools sequential connect
- FastAPI — /analyze endpoint
- HTML UI — file upload + results display
- Deploy + Demo GIF + README

DSA (Weekend only)

- 3 problems per weekend = 6 problems

Content

- 2nd channel: 7 shorts/week batch
- Instagram: 4-5 posts/week
- Main channel: "AI Agent jo resume analyze karta hai" — **upload** (Week 13) 📺
- **LinkedIn pe project post + 5 internship applications**

✓ Checklist

- ☐ PROJECT 2 deployed ✓
- ☐ LangGraph multi-step workflow working
- ☐ FastAPI + HTML UI
- ☐ LinkedIn post + 5 applications sent
- ☐ **Month 3 total DSA: ~52 problems** ✓

📊 Month 3 End

- LangChain Agents — Strong ✓
- LangGraph — Intermediate ✓
- Tool Design — Strong ✓
- DSA — ~52 problems total ✓
- Live Projects — 2 ✓
- Applications sent — 5-10 ✓
- Main channel videos — 6 total ✓
- **Job Readiness — 65%**

20 Oct - 27 Nov

Theme: NLP + Agentic AI Combo + Polish + Interview Ready

WEEK 14 · 20-26 Oct — Docker + NLP Advanced

Resource: *Docker Docs "Get Started" Parts 1-3 + TechWorld with Nana YouTube + HuggingFace Docs*

Tech Topics

- Dockerfile likhna — Python FastAPI app ke liye
- `docker build + docker run + docker-compose.yml`
- `.dockerignore` + env variables in Docker
- Project 1 + Project 2 dockerize karo
- HuggingFace Advanced — AutoTokenizer, AutoModel (BERT family)
- Fine-tuning concept — practically zero se samjho (math nahi, sirf API)

DSA (Weekend only)

- Valid Parentheses Stack, Sliding Window Maximum, Longest Substring Without Repeating, Meeting Rooms II, Task Scheduler

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: Script — "Docker kya hai — AI apps ke liye"

🎯 Target: 5 problems

✓ Checklist

- ☐ Both projects dockerized
- ☐ Docker basics samjha
- ☐ HuggingFace BERT family practically try kiya
- ☐ 5 DSA problems

Resource: NewsAPI (newsapi.org free) + HuggingFace ProsusAI/finbert + LangChain

Tech Topics

- NewsAPI setup — stock ke liye news fetch karna
- HuggingFace pipeline — ProsusAI/finbert (Financial NLP model)
- Sentiment aggregation — Positive/Negative → Buy/Sell/Hold signal
- LangChain agent — news fetch tool + NLP sentiment tool combined
- FastAPI — GET /analyze/{stock} endpoint
- Simple HTML dashboard — stock input + sentiment result card
- Deploy + README + Demo GIF

DSA (Weekend only)

- Find Median from Data Stream, Top K Frequent Elements, Binary Tree Level Order, Zigzag Traversal, Serialize Binary Tree

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: "Docker kya hai" — shoot + edit + **upload** 📺
- **LinkedIn pe Project 3 post karo**

🎯 Target: 5 problems

✓ Checklist

- ☐ PROJECT 3 deployed ✓ (NLP + Agentic AI combo)
- ☐ FinBERT sentiment working
- ☐ LangChain agent + FastAPI
- ☐ LinkedIn post kiya
- ☐ 5 DSA problems

Resource: *Overleaf resume template + LinkedIn optimization*

Tech Topics (Interview Prep)

- ATS-friendly resume — 1 page: ABSA + 3 projects + Skills + Education
- Resume mein impact numbers — "Built RAG app serving multi-PDF queries"
- LinkedIn Headline: "AI Engineer | NLP | Agentic AI | LangChain | HuggingFace"
- LinkedIn pe teeno projects — live URLs ke saath
- Cold email template — AI startups ke liye

DSA (Weekend only)

- Revision: weak topics

Content

- 2nd channel: 7 shorts batch
- Instagram: 4-5 posts
- Main channel: "Stock market AI agent banaya" — shoot + edit + **upload** 📺
- **20+ internship applications bhejo** (LinkedIn, Internshala, AngelList, direct email)

🎯 Target: 3 problems

🔗 Interview Questions Practice

1. RAG kya hai, kaise kaam karta hai?
2. LangChain vs LangGraph — kab kya?
3. Vector DB + similarity search kaise hoti hai?
4. NLP mein embeddings kya hoti hain?
5. FinBERT kya hai, tumne kaise use kiya?
6. ABSA project explain karo
7. Project 1 explain karo
8. Project 2 explain karo
9. Agent kya hota hai, ReAct pattern?
10. NLP + Agentic AI mein difference + overlap?

✓ Checklist

- ☐ Resume ready ✓
- ☐ LinkedIn optimized ✓
- ☐ 20+ applications sent ✓
- ☐ 10 interview questions practice kiye
- ☐ Main channel video 8 uploaded

Resource: *Khud ka GitHub + projects*

Tech Topics

- Teeno projects deploy + test — links working?
- System design: RAG system ka diagram draw karo
- NLP + Agentic AI combined explanation ready karo — resume story
- Mock interview — khud explain karo (record karo, dekho)
- DSA revision — weak topics

DSA (Weekend only)

- Revision mode — 3 problems per weekend = 6 problems

Content

- 2nd channel: 7 shorts/week batch
- Instagram: 4-5 posts/week
- Main channel: "Job search journey — 30 applications in 2 weeks" (Week 17) 📺
- Main channel: "NLP + AI Engineer banne ka safar" — final video (Week 18)
- **10 more applications — total 30+**

✓ Final Checklist

- ☐ DSA total: 70-80 problems ✓
- ☐ 3 projects live + confidently explain ✓
- ☐ ABSA + NLP story clearly explain kar sako ✓
- ☐ 30+ applications sent ✓
- ☐ 2-3 interview calls received ✓
- ☐ Resume + LinkedIn strong ✓

🏠 Final Report (27 Nov 2026)

- Python — Intermediate-Strong ✓
- NLP Practical — Intermediate ✓
- ABSA — Owned + Explained ✓
- LLM APIs — Confident ✓
- LangChain + RAG — Strong ✓
- LangGraph Agents — Intermediate ✓
- Docker — Basic ✓
- DSA — 70-80 problems ✓
- Live Projects — 3 deployed ✓
- Applications — 30+ ✓
- Main channel videos — 10 videos ✓
- **Job Readiness — 80% ✓**

🕒 Resume Story (Nov 2026)

HOW IT ALL STACKS UP

ABSA Project (NLP Research — BERT based)

+

Study Assistant RAG App (LangChain + ChromaDB)

+

Job Application Agent (LangGraph + Agentic AI)

+

Stock News Sentiment Agent (NLP + Agentic AI COMBO)



"NLP + Agentic AI Engineer"

🕒 10-15 LPA target — REALISTIC ✓

👤 Content Summary (2 hrs/day parallel)

Month	Main Channel Videos	2nd Channel	Instagram
Month 1	Python journey + NLP kya hai (2 videos)	28 shorts	~60 posts
Month 2	LangChain + RAG app (2 videos)	28 shorts	~60 posts
Month 3	AI Agent + Job Application Agent (2 videos)	28 shorts	~60 posts
Month 4	Docker + Stock Agent + Job Journey + NLP story (4 videos)	35 shorts	~75 posts
TOTAL	10 main videos	~119 shorts	~255 posts

⚠️ Hard Rules

1. **Tutorial dekho → khud banao** (copy mat karo)
2. **Roz GitHub commit** — streak maintain karo
3. **DSA weekend pe skip mat karo** — minimum 3 problems
4. **Project incomplete mat chordo** — deploy karo tabhi next
5. **NLP + Agentic AI = teri identity on resume** — isko own karo
5. **Content = jo seekha wahi dikhao** — ek kaam teen fayde 🙌